



A two-dimensional coupled hydro-mechanical finite-discrete model considering porous media flow for simulating hydraulic fracturing

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ABSTRACT

In this study, we propose a new coupled hydro-mechanical model considering porous media flow (FDEM-flow) for simulating hydraulic fracturing, which makes full use of the unique topological connection between joint elements and solid elements in the combined finite-discrete element method (FDEM). The joint elements form the flow channel for fluid, through which flow obeys the cubic law. In addition, viscous forces of fluid are taken into account in FDEM-flow. A simple example with analytical solution is given to verify the model. Then, the effects of fluid viscosity on hydraulic fracturing are investigated by this model. The simulated results show that when a low viscosity fluid is injected, the fluid infiltrates the fracture rapidly. The initiation pressure is lower than the theoretical value and the breakdown pressure is slightly larger than the initiation pressure. When high viscosity fluid is injected, fluid infiltrate into the cracks very slowly. The initiation pressure is close to the theoretical value and the breakdown pressure is much larger than the initiation pressure.

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1. Introduction

Hydraulic fracturing is a core technology used in oil and gas wells to increase production, and has been widely used in low-permeability reservoirs and shale gas exploration. It is essentially a rock rupture phenomenon involving hydro-mechanical coupling, which covers a wide range of issues. The study of hydraulic fracturing is of great importance in both theory and practice.

Some hydraulic fracturing models were developed to assist operation design. For example, Perkins and Kern¹ contributed the first model, which was further developed by Nordgren.² Geertsma and Klerk³ developed KGD model. Later, Fung et al.⁴ and Cleary et al.⁵ proposed some semi-analytical methods to solve hydraulic fracturing problems. Other models include the pseudo-3D (P3D) and planar-3D (PL3D) models.^{6,7} More examples of the analytical model can be found in.^{8–15}

Although these analytical models are widely used, a common limitation of which is that they can only be used in simulating the propagation of fractures with highly idealised geometries. Therefore, many numerical methods are used to study hydraulic fracturing. These numerical method can be roughly classified into two

kinds: (1) continuum-based methods; (2) discontinuum-based methods.

A representative in the continuum-based methods is the seepage-stress-damage coupled model.^{16–18} Later, the model was extended to three dimensions by Li et al.,¹⁹ which indicated that the initiation pressure does not coincide with the breakdown pressure. Pan et al.²⁰ combined rock discontinuum cellular automata with TOUGH code to realise a coupling analysis of solid deformable and fluid flow in fractures. Keshavarzi and Mohammadi,²¹ and Chen²² simulated hydraulic fracturing using an extended finite element method. Wu et al.²³ extends the numerical manifold method (NMM) to include the hydro-mechanical model to simulate crack initiation, propagation, block formation and sliding due to water effect. Xie et al.,²⁴ Zhang et al.²⁵ and Zhang et al.²⁶ used displacement discontinuity method to simulate of hydraulic fracturing and its interactions with a pre-existing fracture. Carrier and Granet,²⁷ and Chen²⁸ used finite element method to modelling fluid-driven fracture in permeable medium.

In the discontinuum-based methods, the traditional discrete element method can incorporate joints explicitly. Moreover, the existing commercial codes, such as UDEC²⁹ or 3DEC,³⁰ can deal with multi-physics coupling problems. For example, Nasehi et al.³¹ investigated with UDEC the effect of *in situ* stresses and strength parameters on hydraulic fracturing. However, these methods cannot directly simulate rock mass fractures, difficulties remain in dealing with those rock fractures driven by fluid.

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To simulate the rupture of rock mass, Cundall³² proposed the particle flow method (PFC2D, PFC3D), which provides a promising approach to solving rock fracture driven by fluid flow. For example, Al-Busaidi,³³ Zhao,³⁴ Shimizu,³⁵ Han,³⁶ and Wang³⁷ used the method to study hydraulic fracturing and acoustic emission, and achieved good results. However, the particle flow method has a problem in that the continuum is represented by circular particles bound to each other regardless of how the particles are arranged, the initial model has voids. Therefore, it cannot intuitively characterise the flow paths and crack extension. In addition, the circular particles are rigid, and thus the concept of stress and strain in continuum cannot be directly expressed, but indirectly expressed by statistical averaging with the measurement circle. The stress and strain depend on the radius of the measured circle, the physical meaning of which is not very clear. Ben^{38,39} and Jiao et al.⁴⁰, and Choo,⁴¹ and Morgan⁴² not only considered coupled seepage and stress based on DDA, but also could deal with rock rupture. In addition, Fu et al.⁴³ proposed an explicitly-coupled hydro-geo-mechanical model for simulating hydraulic fracturing. Grasselli et al.⁴⁴ studied the effect of discontinuity on hydraulic fracturing but without considering fluid flow in the fractures.

Yan et al.⁴⁵ based on the combined finite-discrete element method (FDEM) proposed a hydro-mechanical model to simulation of hydraulic fracturing with arbitrarily complex network. In this model, rock matrix is considered to be impervious and pore seepage in rock matrix cannot be considered. However, most reservoir rocks are porous medium through which fluid can flow.⁴⁶ The experimental study carried out by Haimson and Fairhurst^{47,48} and Medlin and Masse⁴⁹ have demonstrated that the porosity and pore fluid have an influence on the hole breakdown pressure. Therefore, in this paper we present a new coupled hydro-mechanical model (FDEM-flow) based on FDEM, which considers both porous media flow and fracture seepage. At the same time, a calibration approach for rock macroscopic permeability is also recommended. Furthermore, we also consider the viscous forces of the fluid.

This paper is organised as follows: first, the basic theory of the coupled hydro-mechanical model considering porous media flow and fracture seepage is introduced; secondly, a simple example with analytical solution is given to verify the model; finally, two hydraulic fracturing problems using different viscosities are studied to examine the influence of pore pressure and fluid viscosity on hydraulic fracturing.

2. Basic assumptions

As shown in Fig. 1, triangular solid elements adjoined through joint elements are used to represent the rock in FDEM,^{50–52} with

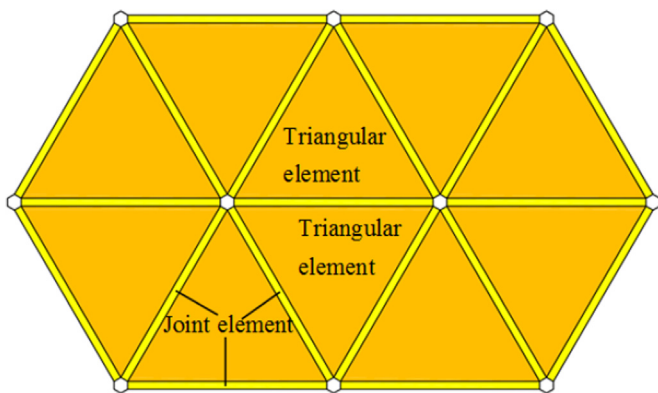


Fig. 1. Connection of triangular elements and joint elements.

the assumption that the solid elements are impervious and all joint elements are allowed bi-directional flow by fluid. Besides an initial aperture a_0 of joint elements is given to represent the permeability of rock matrix.

When some joint elements break (i.e. new cracks generated), the aperture of those joint elements increases resulting in an increased permeability. As a consequence, both seepage in fracture and porous media flow in rock matrix can be characterised.

3. Considerations of rock permeability in FDEM-flow

Based on the above assumptions, we will see that problems of both porous media flow and fracture seepage can be reduced to pure fracture seepage problems. The hydro-mechanical model includes the following process: fluid flow through a crack applies pressure to the two walls of the crack, resulting in opening or closing under the action thereof. However, opening or closing of the crack will affect the fluid flow in the crack and lead to a fluid pressure change.

As shown in Fig. 2(a), fluid in the crack applies fluid pressure onto its two walls. Under the action of fluid pressure, the crack aperture might increase if the fluid pressure is large enough (see Fig. 2(b)). According to the cubic law, i.e. the flow rate is proportional to the cube of the aperture, crack opening affects the flow rate and fluid pressure in turn.

Since joint elements are inserted in the common edge of each triangular element in FDEM, triangle solid elements and joint elements form a topology as shown in Fig. 3. Only the joint elements sustain fluid flow, and all joint elements will be involved in the seepage calculation. In Fig. 3, a white node denotes an intersection of the fluid flow network paths, and also an intersection of joint elements and triangular elements. A segment (i.e. a joint element) between any two adjacent white nodes constitutes a flow path, such as the path connecting Nodes 4 and 2. While all the joint elements in one dashed circle with a node as the centre constitutes the node control volume. The control volume of a node is constituted by all the joint elements in one dashed circle surrounding the node. In the seepage calculation, the fluid pressure at a node will represent the fluid pressure of the dashed circle.

The procedure to solve the flow rate between the two nodes is explained below. The pressure difference between two nodes is first calculated. Take Node 2 in Fig. 3 as an example. The total pressure difference between Node 2 and Node 1 is given by

$$\Delta p = p_2 - p_1 + \rho_w g(y_2 - y_1), \quad (1)$$

where ρ_w is density of fluid, p_1 and p_2 are fluid pressure at Nodes 1 and 2, and y_1 and y_2 are the vertical coordinates of Nodes 1 and 2, respectively.

Using the cubic law, the flow rate from Node 1 to Node 2 is calculated as

$$q_{21} = -\frac{1}{12\mu} a^3 \frac{\Delta p}{L}, \quad (2)$$

where, a is the average aperture of the joint element, μ is the fluid viscosity, and L is the length of the joint element. The introduction of the negative sign is due to the convention that flow-out is negative and flow-in is positive.

The average aperture a is not constant, which depends on the average normal displacement of the joint element as illustrated in Fig. 4(a). The average aperture of the joint elements is expressed as

$$a = a_0 + u_n, \quad (3)$$

where a_0 represents the initial aperture of the joint element, and u_n is the average normal displacement of the joint element

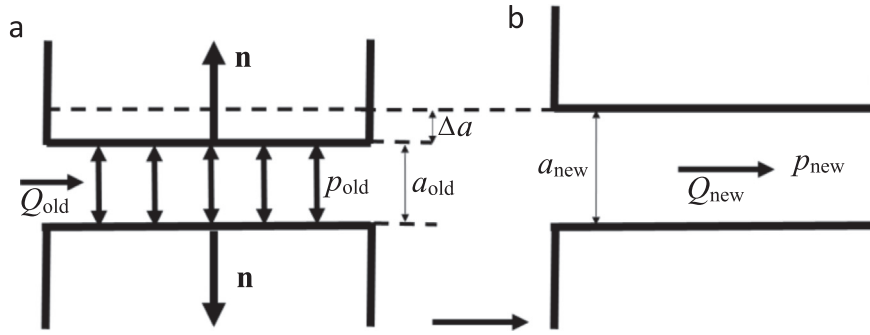


Fig. 2. Fluid flow and crack deformation: (a) the effect of fluid pressure on fracture; and (b) the effect of crack aperture on fluid flow and pressure.

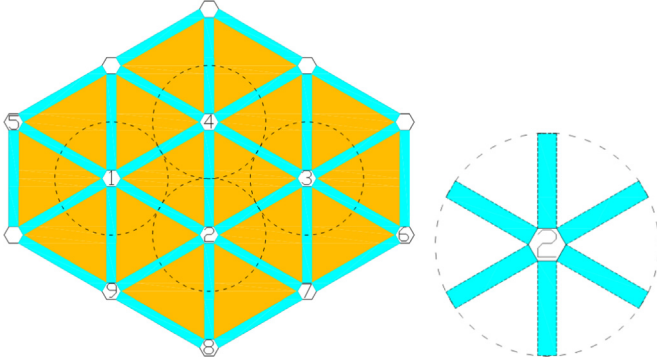


Fig. 3. Fluid flow in the joint element network.

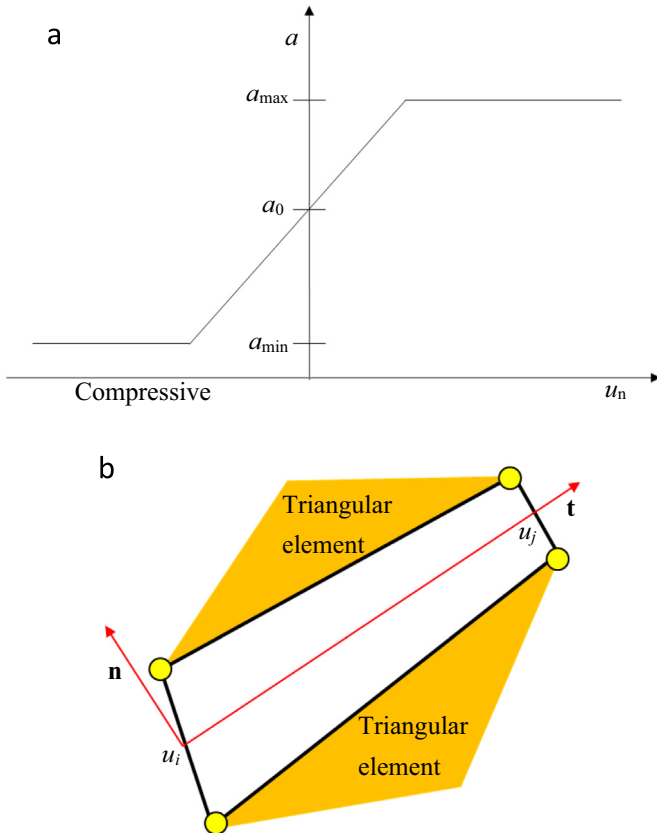


Fig. 4. (a) Relationship between hydraulic aperture width and average normal joint element displacement; and (b) average normal displacement of joint element.

(positive denoting opening), which can be obtained from the average normal displacement of two end points of the joint element, i.e. $u_n = (u_i + u_j)/2.0$, as shown in Fig. 4(b). u_i and u_j are

projections of the displacement of two endpoints in the normal direction, respectively.

When conducting seepage calculations, the aperture of a joint element has a minimum aperture $a_{\min}$ as the lower bound. If the aperture calculated by Eq. (3) is less than $a_{\min}$, it will be set to $a_{\min}$. To describe the lowered permeability of the rock, $a_{\min}$ can be set to a small value.

It should be noted that, according to Eq. (2), the flow between the two nodes still occurs due to the effect of gravity even if the fluid pressure at Node 1 and 2 are both zero, which is deemed unreasonable. The valid case is that the flow rate entering Node 2 should decrease as the saturation of Node 1 declines. For example, if the saturation of Node 1 is zero, the flow rate entering Node 2 from Node 1 is also zero. Therefore, the flow rates computed by Eq. (2) should be multiplied by a coefficient f_s , which is a function of the degree of saturation s :

$$f_s = s^2(3 - 2s) \quad (4)$$

This empirical function has the following characteristics: $f_s = 0$ if $s = 0$, $f_s = 1$, if $s = 1$, and $0 \leq f_s \leq 1$ for any $s \in [0, 1]$. That is to say, the actual flow rate entering Node 2 from Node 1 will decrease if Node 1 is unsaturated. Moreover, the derivative of the function at $s = 0, 1$ is zero. Therefore, the flow rates computed by Eq. (2) are multiplied by the coefficient f_s , and flow will not occur in nodes with zero saturation. Thus, the modified flow rates between two nodes are,

$$q_{21} = -k_j a^3 \frac{\Delta p}{L} f_s. \quad (5)$$

As shown in Fig. 3, since Node 2 is also connected with Nodes 3, 4, 7, 8, and 9, the flow rate from those nodes can be computed in a similar manner. There, the total flow rate at Node 2 can be calculated as $Q = q_{21} + q_{23} + q_{24} + q_{27} + q_{28} + q_{29}$. Accordingly, the current pressure at Node 2 can be obtained as²⁹

$$p = p_0 + K_w Q \frac{\Delta t}{V} - K_w \frac{\Delta V}{V_m}, \quad (6)$$

where p_0 is the pressure at Node 2 at the preceding time step, K_w is the bulk modulus of fluid, Q is the total flow rate at Node 2, Δt is the time step, $\Delta V = V - V_0$, and $V_m = \frac{(V + V_0)}{2}$. Here, V and V_0 are the volumes of Node 2 at the current time step and preceding time step, respectively. The volume of a node is defined as half of the total volume of all joint elements connected to the node. Take Node 2 in Fig. 3 as an example, the volume of Node 2 is defined as $V = (V_{j21} + V_{j23} + V_{j24} + V_{j27} + V_{j28} + V_{j29})/2$.

If $p > 0$ is derived from Eq. (6), the change in volume of Node 2 leads to a pressure change of it. Otherwise, if $p < 0$, it is indicated that the fluid in the volume of Node 2 is insufficient to fill that volume. In this case, we set $p = 0$ and update the saturation of Node 2 to

$$s = s_0 + Q \frac{\Delta t}{V} - \frac{\Delta V}{V_m} \quad (7)$$

where s_0 is the saturation of Node 2 at the preceding time step. It should be indicated that if $s < 1$, p is set to zero and Eq. (7) is used instead of Eq. (6). If $s > 1$, we set $s = 1$ and Eq. (6) is applied.

To ensure the numerical stability of the seepage calculation, the time step should not exceed the following thresholds

$$\Delta t_f = \min \left[\frac{V}{K_w \sum_i \frac{a_i^3}{12\mu}} \right] \quad (8)$$

where a_i hydraulic aperture of the i th joint element connecting to the node.

4. Calibration of the initial aperture of joint elements

Since the initial aperture a_0 of joint elements associated with a mesh of the problem domain cannot be measured directly, this section will develop a systematic way to calibrate a_0 such that the mesh with a_0 as the initial aperture of joint elements has the same permeability as the actual rock in study.

Firstly, a rock sample is taken for the permeability test. The rock sample has a width of W , a height of H and a unit thickness. Prior to the test, the rock sample is made fully saturated. The top and bottom boundary of the sample is set impermeable. Then, fluid flow through the sample is established by maintaining a constant fluid pressure P_{in} on the left-hand side of the sample and another constant fluid pressure P_{out} on the right-hand side ($P_{in} > P_{out}$). When the steady flow state is achieved, we have $Q = Q_{in} = Q_{out}$. Here, Q_{in} = inflow, Q_{out} = outflow; and Q = flux through the sample in the steady flow state.

Under the assumption that the rock sample is isotropic, the steady flow rate Q according to Darcy's law is given as³⁵

$$Q = \frac{kH(P_{in} - P_{out})}{\mu W}, \quad (9)$$

where μ and k are the fluid viscosity and the macroscopic permeability of the rock sample, respectively. The permeability of the rock sample is thus given by

$$k = \frac{Q\mu W}{H(P_{in} - P_{out})}. \quad (10)$$

Now the rock sample is discretized with the same mesh density as the actual mesh for the problem domain. For the given fluid and

the given mesh of the sample such as Fig. 7, Q is uniquely determined by the initial joint element aperture a_0 , and can be calculated through the algorithm stated in the preceding section. According to Eq. (10), k is a function of a_0 , denoted by $k(a_0)$. As a consequence, the proper a_0 is such that $k(a_0) = \bar{k}$, with $\bar{k}$ being the actually measured permeability of the rock in study.

5. Fluid pressure and fluid viscous forces

In FDEM-flow, a linearly distributed fluid pressure is assumed as shown in Fig. 5(a), with the fluid pressure at Node 1 being p_1 and the fluid pressure at Node 2 being p_2 . Thus the total normal fluid pressure acting on the two sides of the triangular elements connected by the intermediate joint element can be given by

$$f_n = \frac{(p_1 + p_2)}{2} L, \quad (11)$$

where L is the length of the joint element, as shown in Fig. 5(a). In addition to the fluid pressure in the normal direction, we also take into account the tangential viscous force in FDEM-flow.

According to the fluid flow algorithm in³⁵ the fluid flow in the joint elements is assumed to be a two-dimensional Poiseuille flow, i.e. parallel plate laminar flow. Therefore, the laminar flow between two parallel edges of two triangular elements extending in the x -direction, as shown in Fig. 5(b) will now be considered. The edges lie in the planes $r = \pm a/2$, and the flow is in the tangential direction, with no velocity component in the normal direction. The velocity distribution for laminar flow between parallel edges is a function of r only, as shown in Fig. 5(b), defined by

$$v = \frac{1}{2\mu} \left[\left(\frac{a}{2} \right)^2 - r^2 \right] \frac{\Delta p}{L}. \quad (12)$$

Viscous fluid flow along a solid boundary will induce a shear stress on the boundary. The shear stress at a surface parallel to a plate, at point r , is given by

$$\tau = -\mu \frac{dv}{dr} = r \frac{\Delta p}{L}. \quad (13)$$

In particular, at the parallel edges of two triangular elements ($r = a/2$), the shear stress is given by³⁵

$$\tau_{f0} = \frac{a}{2} \frac{\Delta p}{L}. \quad (14)$$

Thus, the total tangential viscous force acting on the edges of the two triangular elements is given by

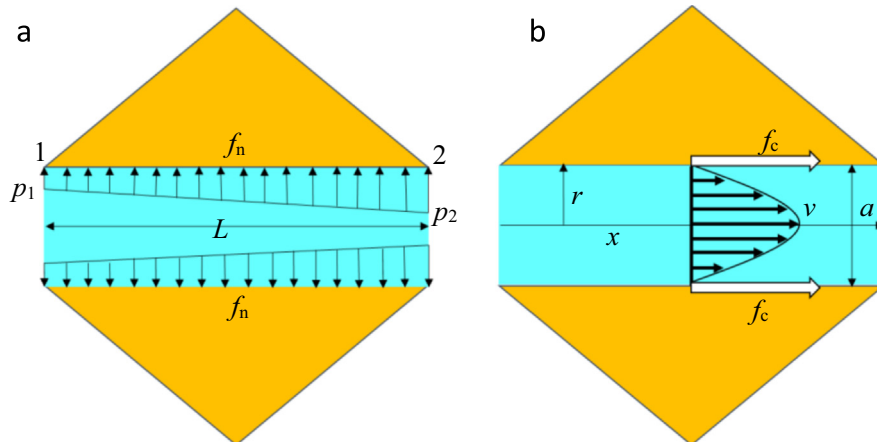


Fig. 5. Fluid flow and forces in the joint element: (a) fluid pressure in a joint element; and (b) fluid tangential velocity distribution and the viscous force in a joint element.

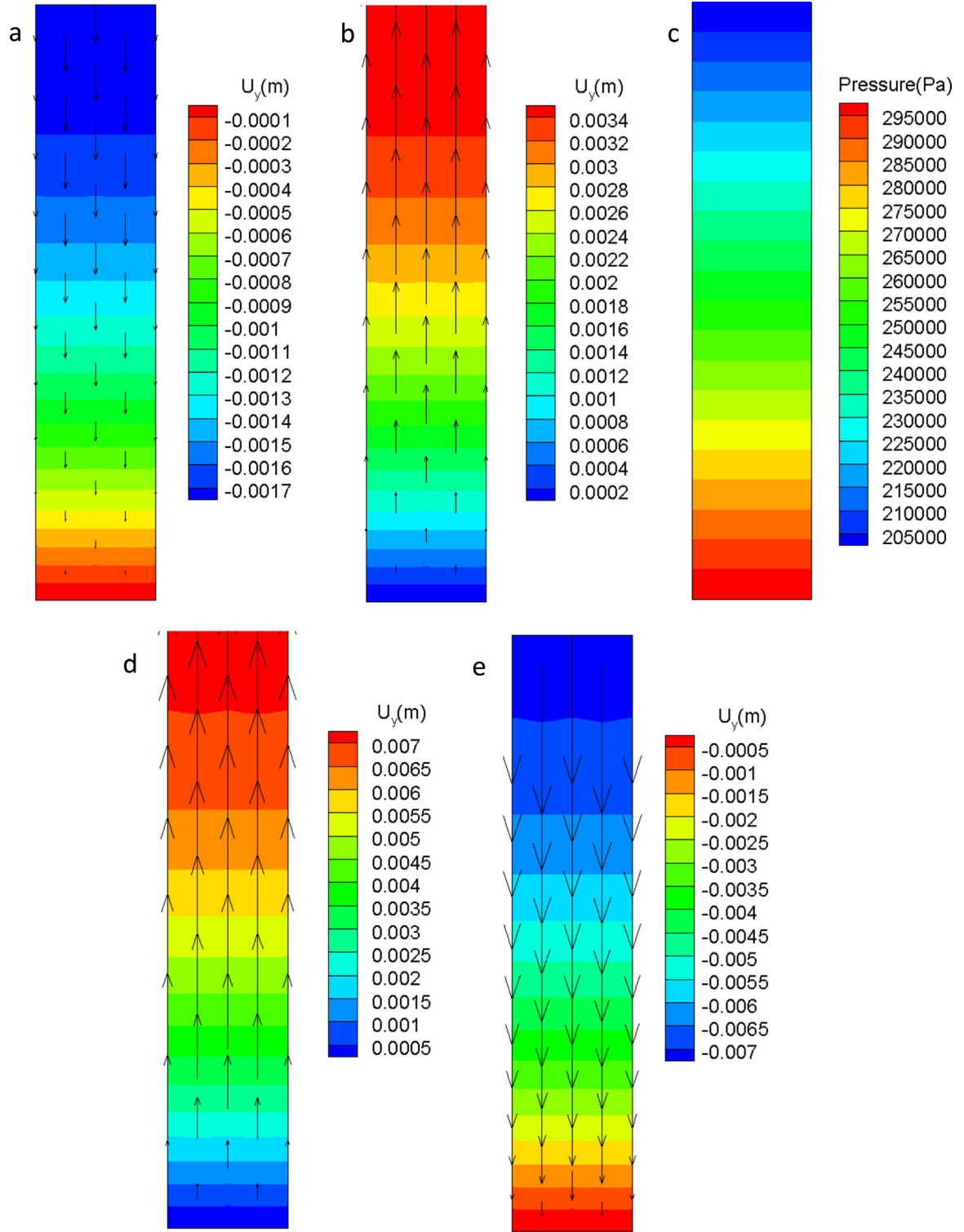


Fig. 6. The simulation results by the coupled hydro-mechanical model.

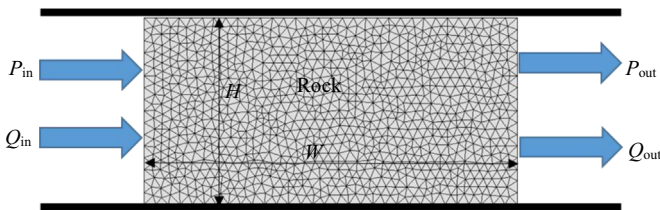


Fig. 7. Simulation of permeability test.

$$f_c = L\tau_{f_0} = \frac{a}{2}\Delta p. \quad (15)$$

As shown in Fig. 5(b), the total viscous force f_c is applied to the edges of two triangular elements that form the channels. Thus, by introducing the above fluid flow algorithm, both the fluid pressure and viscous forces on the triangular element are considered.

After the calculation of the fluid pressure and tangential viscous force in each joint element, called the fluid force uniformly, as

with the contact force and external load, the fluid force, as a distributed load acting on the corresponding edges of the triangular solid elements, is also assigned to each node of the solid elements.

6. Calculation procedure

The preceding sections mainly focus on how to perform the seepage calculation, and how to apply the fluid forces. In this section, we describe FDEM-flow from a holistic perspective that combines mechanics and seepage flow. The entire calculation process is as follows. Firstly, calculate the nodal forces based on solid and joint element deformation at the current time step. Secondly, calculate the nodal forces caused by the contact force. Thirdly, conduct the seepage calculation to update the fluid pressure and viscous force at each node. Finally, update the velocity and coordinates of the nodes according to Newton's second law. This is a typical time step. All the time steps are conducted till the final time step.

7. Verification of the coupled hydro-mechanical model

A rectangular strip with height $H=10$ m and width $W=2$ m, the bottom, left and right boundaries of it are applied normal constraints. Calculation parameters include: elastic modulus $E=81.818$ MPa, Poisson's ratio $\nu=0.364$, solid density $\rho_d=500$ kg/m³, the bulk modulus of fluid is 2.2 GPa, water density $\rho_w=1000$ kg/m³, gravitational acceleration is set to 10 m/s². (1) Initially, no fluid action, find the displacement of model at the top under the action of gravity; (2) then the pore pressure is fixed at zero at the top of the model, and the saturation is initialised to 1 and fixed at the top. Find the incremental displacement of the model at the top boundary; (3) continue from stage (2), and the water table is raised to 20 m above the top of the model. Find the incremental displacement of the model at the top; (4) continue from the previous stage, and apply a pore pressure of 0.5 MPa at the base. Find the incremental displacement of the model at the top; (5) continue from the stage (3), and apply a pore pressure of 0.1 MPa at the base. Find the incremental displacement of the model at the top. Since the above problems have analytical solution,⁵³ we use them to verify the coupled hydro-mechanical model in this paper.

At stage (1), the analytical solution for the vertical incremental displacement of the model at top is given by

$$u_y = -\frac{\rho_d g H^2 (1 + \nu)(1 - 2\nu)}{2E(1 - \nu)} \quad (16)$$

where ρ_d is density of solid, E , ν are elastic modulus and Poisson's ratio, respectively. The calculated value from the coupled hydro-mechanical model is $-1.786\text{e-}3\text{m}$, which matches the analytical value at this stage.

At stage (2), the analytical solution for the vertical displacement of the model at top is given by:

$$u_y = \frac{\rho_w g H^2 (1 + \nu)(1 - 2\nu)}{2E(1 - \nu)} \quad (17)$$

The calculated value from the coupled hydro-mechanical model is $+3.572\text{e-}3\text{m}$, which matches the analytical value at this stage. The incremental displacement distribution is as shown in Fig. 6(b).

At stage (3) the analytical solution for the vertical incremental displacement of the model at top is zero and the analytical solution for hydrostatic pressure distribution is given by

Table 1

Equivalent elastic parameters and flow parameters.

Parameter	Value
Rock	
Bulk density, ρ (kg/m ³)	2600
Young's modulus, E (GPa)	55
Poisson's ratio, ν	0.25
Tensile strength, f_t (MPa)	6
Internal cohesion, c (MPa)	20
Friction angle of intact material, ϕ (°)	30
Permeability	4×10^{-16} m ²
Fluid flow parameters	
Bulk modulus of the fluid, K_w (GPa)	2.2
Viscosity of the low viscosity fluid, μ (mPa s)	0.1
Viscosity of the high viscosity fluid, μ (mPa s)	100
Density of fluid, (kg/m ³)	1000

$$p^{(3)} = p_b^{(3)}(1 - \frac{y}{H}) + p_t^{(3)}\frac{y}{H} \quad (18)$$

where $p_b^{(3)}$ is fluid pressure at the bottom of the model at stage (3), $p_t^{(3)}$ is fluid pressure at the top of the model.

The calculated value from the coupled hydro-mechanical model is $1.4\text{e-}7\text{m}$, which matches the analytical value at this stage. The hydrostatic pressure distribution is as shown in Fig. 6(c).

At stage (4), the analytical solution for the vertical incremental displacement of the model at top is given by:

$$u = \frac{(p_b^{(4)} - p_t^{(3)})(1 + \nu)(1 - 2\nu)}{2E(1 - \nu)}H \quad (19)$$

where $p_b^{(4)}$ is fluid pressure at the bottom of model at stage (4).

The calculated value from the coupled hydro-mechanical model is $7.143\text{e-}3\text{m}$, which matches the analytical value at this stage. The displacement distribution is as shown in Fig. 6(d).

At stage (5), the analytical solution for the vertical incremental displacement of the model at top is given by:

$$u = \frac{(p_b^{(5)} - p_t^{(3)})(1 + \nu)(1 - 2\nu)}{2E(1 - \nu)}H \quad (20)$$

where $p_b^{(5)}$ is fluid pressure at the bottom of the model at stage (5).

The calculated value from the coupled hydro-mechanical model is $-7.143\text{e-}3\text{m}$, which matches the analytical value at this stage. The hydrostatic pressure distribution is as shown in Fig. 6(e).

8. Hydraulic fracturing simulation

8.1. Rock specimen model and loading conditions

A borehole for fluid injection is created at the centre of the rock model with **2 m** both in width and in height. The diameter of the borehole is **0.2 m**. The entire rock model is meshed into 15,303 triangular solid elements, with the joint elements inserted between the solid elements. The **in situ stress with 10 MPa in the x-direction and 5 MPa in the y-direction**, are applied to the rock model.

To consider the effect of fluid viscosity on the hydraulic fracturing operations, a low viscosity fluid (**$\mu=0.1$ mPa s**) and a high viscosity fluid (**$\mu=100$ mPa s**) are used as the fracturing fluid to investigate the influence of fluid viscosity on the hydraulic fracturing behaviour. Fracturing fluid is **injected at 0.0004 m³/s**.

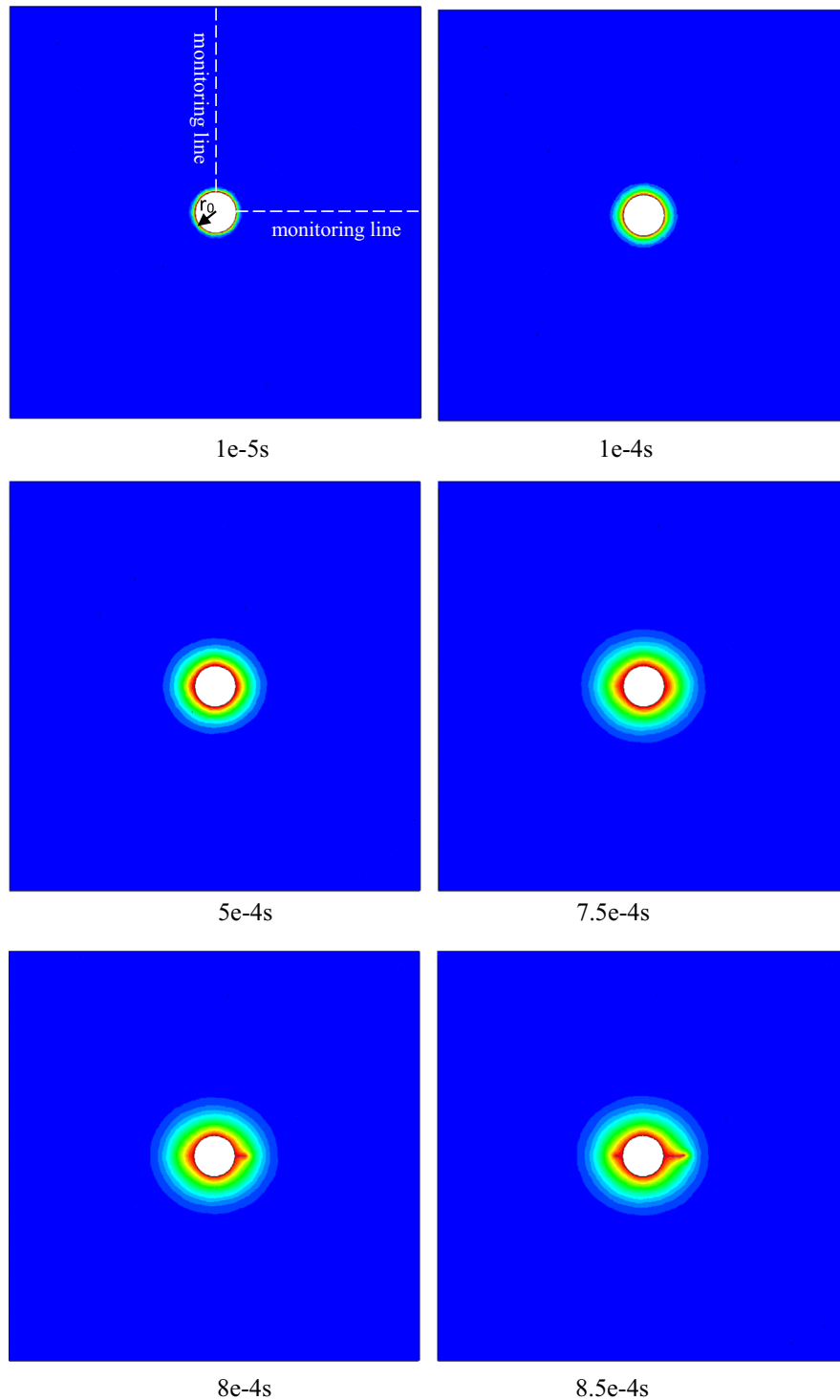


Fig. 8. Low viscosity (0.1 mPa s) fluid injection: fluid pressure distribution and crack propagation.

8.2. Calibration of microscopic parameters

The equivalent elastic parameters and flow parameters are listed in Table 1. Microscopic input parameters for hydraulic fracturing simulation are calibrated by simulating a Brazilian disc test and uniaxial compression strength tests. Here, we focus on how to obtain the flow input microscopic parameters. As mentioned previously, the initial aperture a_0 , which, associated with the mesh of the rock model, determines the permeability of the discrete model, cannot be determined directly, and must be first

calibrated. As shown in Fig. 7, a discrete rock model measuring 1 m in width and 0.5 m in height is used for the permeability test (the size of triangular elements used in the rock model is almost the same as the mesh used in this hydraulic fracturing simulation). We obtain the initial aperture a_0 by making the permeability of the discrete rock model consistent with that of the rock in study. On calibration, the actual rock permeability of $4 \times 10^{-16} \text{ m}^2$ corresponds to $a_0 = 4.17 \times 10^{-6} \text{ m}$ associated with the mesh in Fig. 7.

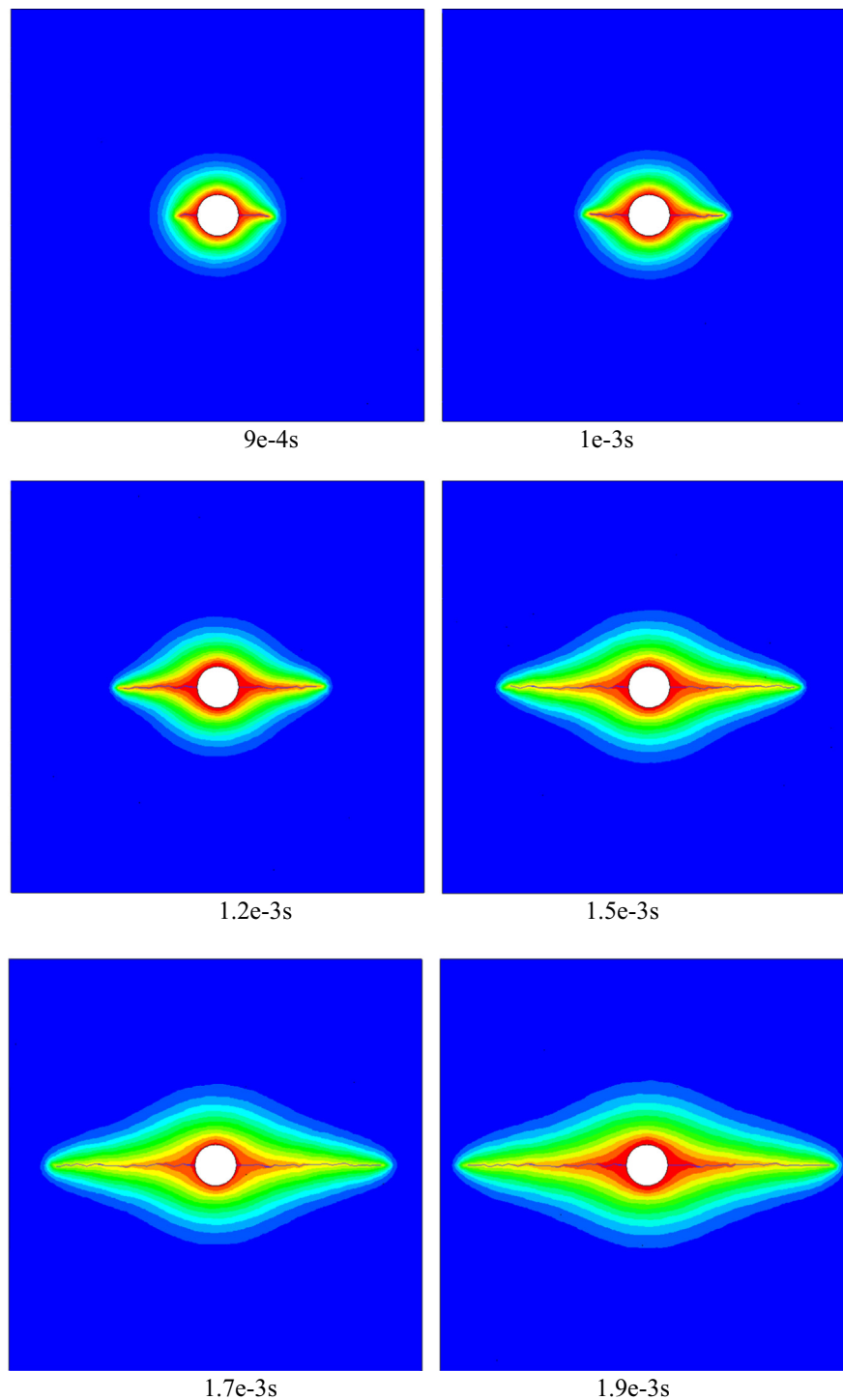


Fig. 8. (continued)

8.3. Simulation results: low viscosity fluid injection

In this section, we present the simulation results from FDEM-flow when a low viscosity fluid ($\mu=0.1$ mPa s) is used. To better analyse and compare the results, we set two monitoring lines in the rock model as shown in Fig. 8 at time 1e-5s to monitor the major principal stress (tension stress positive) and fluid pressure.

Shown in Fig. 8 are fluid pressure distribution and crack propagation at different time, with the low viscosity fluid infiltrated. Before rock fracture, although the rock itself has a very low permeability, yet the permeability coefficient of the joint element is

large due to a small fluid viscosity μ according to Eq. (2). Thus, there is still a quantity of fluid flow into the joint elements, see the cases of 1e-5s to 7.5e-4s in Fig. 8, where no crack is generated, but a fluid pressure is distributed near the borehole and a small elliptical zone is formed. We can see this clearly at 7.5e-4s in Fig. 8: the red zone is an ellipse, indicating that even before rock fracture, the original isotropic permeability starts to become anisotropic under the effects of *in situ* anisotropic stress. The permeability coefficient is large in the direction of the major principal stress because the horizontal confining stress (10 MPa) is larger than the vertical confining stress (5 MPa), which causes the deformation in

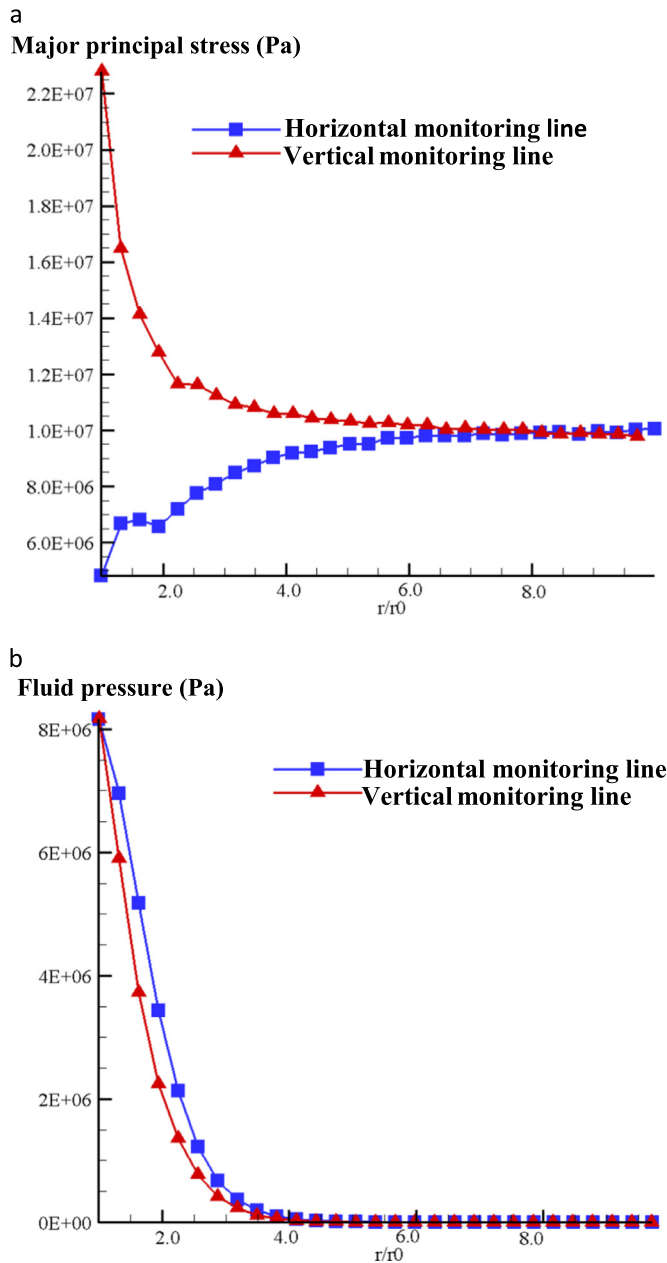


Fig. 9. Major principal stress and fluid pressure distributions, after applying stress, but before injecting fluid: (a) maximum principal stress distribution; and (b) fluid pressure distribution.

vertical direction is easier than that in horizontal direction.

As shown in Fig. 9(a), within $r/r_0 < 6$ (r represents the distance of a monitoring point from the centre of the borehole, and r_0 is the radius of the borehole), when the values of r/r_0 for different monitoring points were the same, the major principal stress on the horizontal monitoring line is smaller than that on the vertical monitoring line (compressive stress). Therefore, the joint elements on the left and right sides near the borehole are easier to open, and the permeability coefficient of these joint elements is larger than those of the top and bottom sides around the borehole, and hence the fluid flows more easily in the horizontal direction. Then the fluid pressure at the left and right sides near the borehole is greater than that at the top and bottom sides. As a consequence, the fluid pressure near the borehole is elliptically distributed.

As shown in Fig. 9(b), within the range $r/r_0 < 4$, under the same value of r/r_0 , the fluid pressure at each horizontal monitoring point

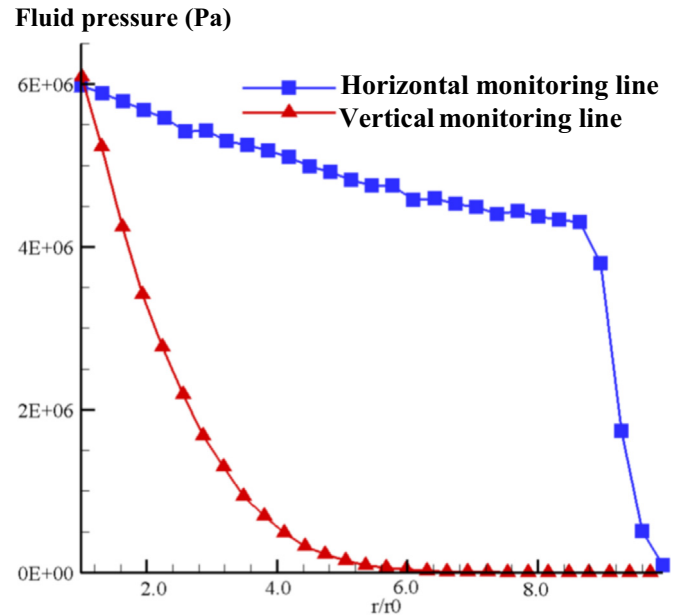


Fig. 10. Fluid pressure distribution at 1.9e-3s.

is larger than that on the vertical monitoring line. This indicates that under the effects of *in situ* anisotropic stress, the permeability coefficient in the horizontal direction is larger than that in the vertical direction. Nevertheless, before rock fracture, the effect of the *in situ* stress on the permeability coefficient is limited. The region subjected to a fluid pressure distribution is also growing in size, as shown in Fig. 8 at 1e-5s to 7.5e-1s.

With time evolution, the injected fluid continues to increase in volume and pressure. Beyond 7.5e-4s, fracture starts at the left side of borehole, as shown in Fig. 8 at time 8e-4s. Thereafter, fracture also comes at the right-hand side of the borehole. Although the fractures on both sides do not start at the same time, as shown in Fig. 8 at 8.5e-4s and 9e-4s, this is mainly due to the asymmetry of the mesh. Ultimately, the cracks on both the left and right become nearly symmetrical and propagate to both sides, as shown in Fig. 8 at 1.2e-3s to 1.9e-3s. In Fig. 8, we can see that cracks extend along the direction of the major principal stress, which is also consistent with the experimental results.⁵⁴ At the same time, once cracks are produced, namely, joint elements are broken, the aperture of each broken joint element increases, resulting in the increase in the permeability coefficient of fracture. Thus, the fluid will flow along the crack, causing the fluid pressure at points in the fracture to be larger than at other points with the same distance from the borehole.

The pressure distribution of the fluid becomes prolate with crack propagation, suggesting that once cracks are produced, the anisotropy of the permeability coefficient of the sample will become stronger. The permeability coefficient along the direction of crack propagation is much larger than the permeability coefficient in the other direction, as shown in Fig. 10. Therefore, after crack generation, the *in situ* stress has a more significant effect on permeability. Alternatively, cracks are wrapped inside the fluid pressure distribution zone, as shown in Fig. 8 at 8e-4s to 1.9e-3s. This also indicates that once cracks extend, the low viscosity fluid will quickly enter newly generated cracks, and exert fluid pressure on the fracture. According to linear elastic fracture mechanics,⁵⁵ it will make subsequent cracking easier.

With the continuous injection of the fluid, as shown in Fig. 11 (a), the fluid pressure inside the borehole increases. But it is not linearly increasing due to the viscosity of fluid. Because some of the fluid permeates into the rock samples, the accumulated

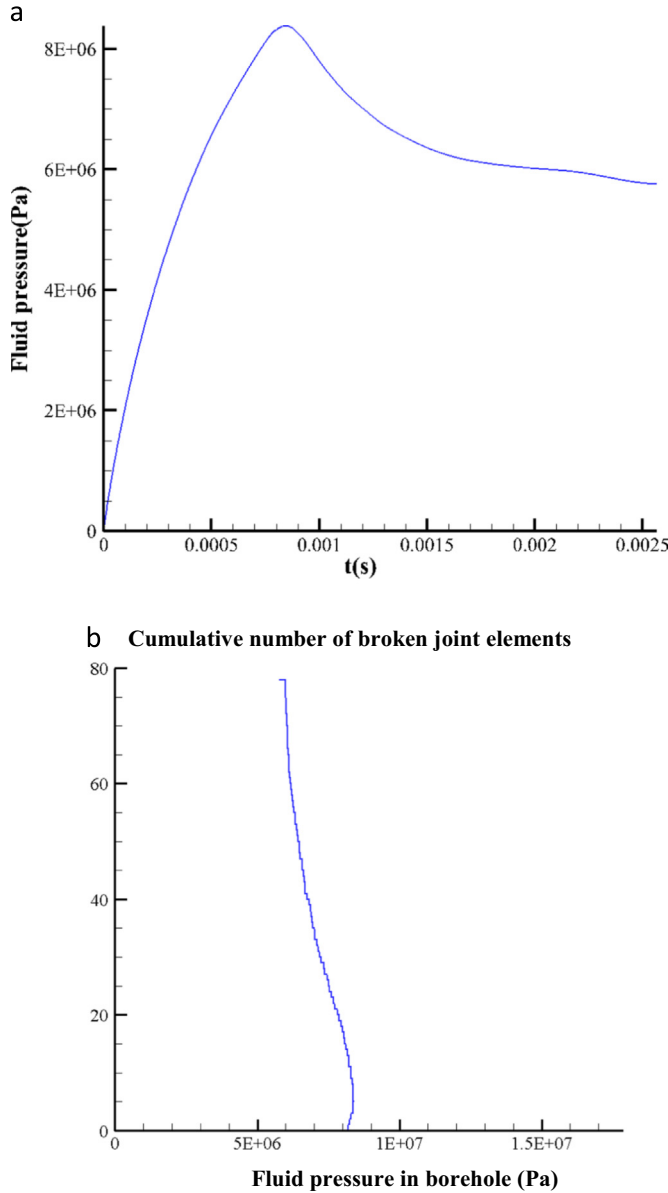


Fig. 11. When a low viscosity fluid is injected: (a) fluid pressure - time step curve; and (b) the cumulative number of broken joints-fluid pressure curve.

amount of fluid within the borehole is not increasing linearly. Despite this, after 8e-4s, the sample still produces cracks (shown in Fig. 8), and the fluid pressure does not decrease, but instead continues to increase. This is because the aperture is still small at the beginning of fracture, the flow rates into newly generated cracks remain less than the injection flow rate, and therefore, the cumulative amount of fluid within the borehole still increases, causing fluid pressure within the borehole continuing to increase until after 1e-3s when the fluid pressure starts to decrease: the fluid pressure at this time corresponds to the breakdown pressure (as shown in Fig. 11(a), the breakdown pressure is 8.4 MPa).

In addition, according to the cumulative number of broken joint elements within the fluid pressure curve, as shown in Fig. 11(b), when the borehole fluid pressure is 8.1 MPa (close to the breakdown pressure of 8.4 MPa), cracks are produced in rock. This is much less than the initiation pressure calculated by theory

$$P_b = 3\sigma_h - \sigma_H + f_t = 11 \text{ MPa}, \quad (21)$$

as in.⁵⁴ This is because the fluid viscosity is very small, the fluid

can seep into the rock around the borehole as shown at 7.5e-4s in Fig. 8. Thus, a pore pressure gradient is generated in there. The direction of the pore pressure gradient is apparently consistent with the subsequent direction of crack propagation. According to^{56,57} fracture extension is promoted while the direction of the pore pressure gradient is consistent with the direction of crack propagation. This indicates that the simulation results are consistent with the experimental results from Bruno.⁵⁶ We can conclude that FDEM-flow can better reflect the effect of pore pressure and pore pressure gradient on crack initiation pressure.

Moreover, another experimental study⁵⁸ shows that, the initiation pressure predicted by traditional theoretical formulae, $P_b = 3\sigma_h - \sigma_H + f_t$ based on linear elastic theory,⁵⁴ is different from the test measured pressure. The reason for this is that the theoretical formula is based on linear elasticity, without considering the effect of pore pressure.

8.4. Simulation results: high viscosity fluid injection

In Section 8.3, the simulation results for a low viscosity fluid injected were analysed to show that FDEM-flow is feasible for modelling rock fracture driven by pressured fluid, and that it can characterise the fracture seepage and permeability of rock. In this section, a highly viscous fluid ($\mu = 100 \text{ mPa s}$) is used: the simulation results are analysed and compared with those from a low viscosity fluid. The same two monitoring lines are set, see 1e-5s in Fig. 12.

As shown in Fig. 12, when a fluid of high viscosity μ is injected and before cracking, the permeability coefficient of joint elements is small for a large fluid viscosity μ according to Eq. (2). Although all joint elements allow fluid to flow before they breaks, yet the fluid is less able to enter the rock because the fluid has a high viscosity. As a result, before cracking, the fluid pressure distribution only focuses on the perimeter of the borehole, and is not expanded. As shown in Fig. 12, from 1e-5s to 4e-4s the fluid pressure distribution keeps constant, the fluid pressure is distributed only on the perimeter of the borehole, and is not an elliptical distribution as a low viscosity fluid was used. This shows that the original isotropic model, when using high viscosity fluid and before cracking, the anisotropy of the permeability is not obvious even under the condition of an anisotropic *in situ* stress field.

As shown in Fig. 13, we draw the fluid pressure distribution along the horizontal monitoring line and the vertical monitoring line, both passing through centre of the borehole (at 4e-4s, before the generation of any fractures), indicating that the fluid pressure distributions are similar. This implies that the anisotropy in the permeability is low before cracking, and the effect of anisotropic *in situ* stress on the permeability coefficient is also small. This is not the case of a low viscosity fluid injection, where the fluid pressure on the horizontal monitoring line is higher than the pressure on the vertical monitoring line (when r/r_0 was the same), see Fig. 10.

When the fluid pressure is increased to a certain value, fracture comes on the left and right of the borehole, as shown at 4.5e-4s and 6e-4s in Fig. 12. With the use of low viscosity fluid injection, in a similar manner, cracks also extend along the direction of the major principal stress, and both the growth paths are the same. This shows that the crack propagation direction is in the direction of the major principal stress, regardless of the viscosity of the fluid. At the same time, it can be found that the fluid pressure distribution only focuses on the borehole perimeter and the cracks, differing from the case of low viscosity fluid injection. Moreover, as shown at 4.5e-4s and subsequent in Fig. 12, fractures are found far from the borehole where no fluid pressure is observed,

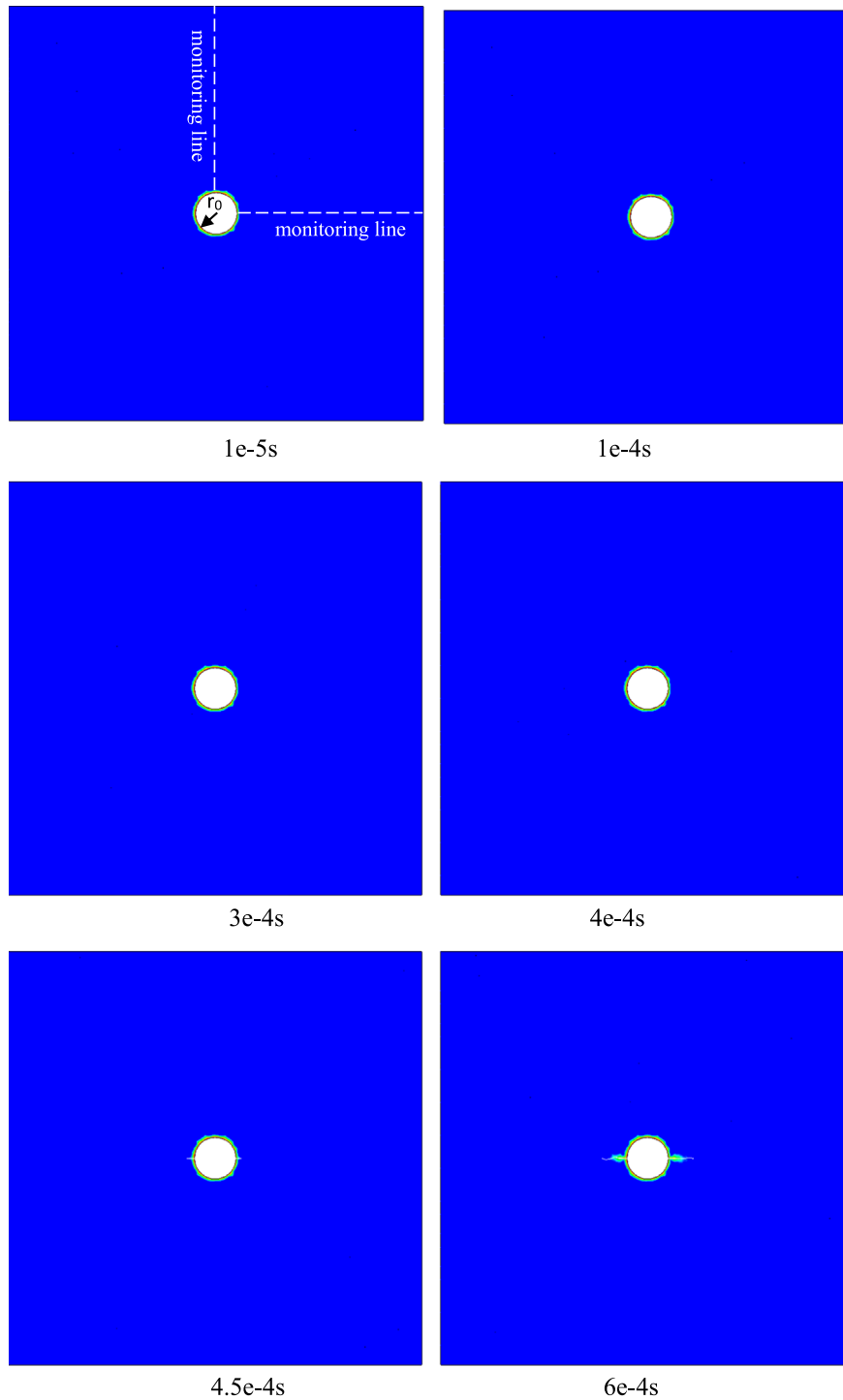


Fig. 12. Fluid pressure distribution and crack propagation: high viscosity fluid used (100 mPa s).

differing from the case of low viscosity fluid injection, where the fluid pressure distribution encompassed the entire crack. This indicates that, even after the cracks are generated, the fluid does not infiltrate into the crack very quickly. As shown in Fig. 14, the fluid pressure at points along the horizontal monitoring line is small at its end. As shown at 8e-4s to 1.3e-3s in Fig. 12, the crack is symmetrically extended along the direction of the major principal stress and the fluid pressure is only linearly distributed along the crack. As shown in Fig. 14, the fluid pressure on the horizontal monitoring line decreases along the fracture, the fluid pressure along the vertical monitoring line quickly decays to zero; when the

radial distance is the same, the fluid pressure along the horizontal monitoring line is greater than that on the vertical (the fluid pressure fluctuation along the horizontal monitoring line is because the selected monitoring points are not just located in cracks). This indicates that the permeability along the direction of fracture was much larger than normal to it. Thus, after cracking, the *in situ* stress has a significant effect on the permeability.

Also, in conjunction with Fig. 15(a), which shows that with the continuous injection of fluid, the fluid pressure inside the borehole increases linearly. This differs from the low viscosity fluid injection where the fluid pressure within the borehole increases non-

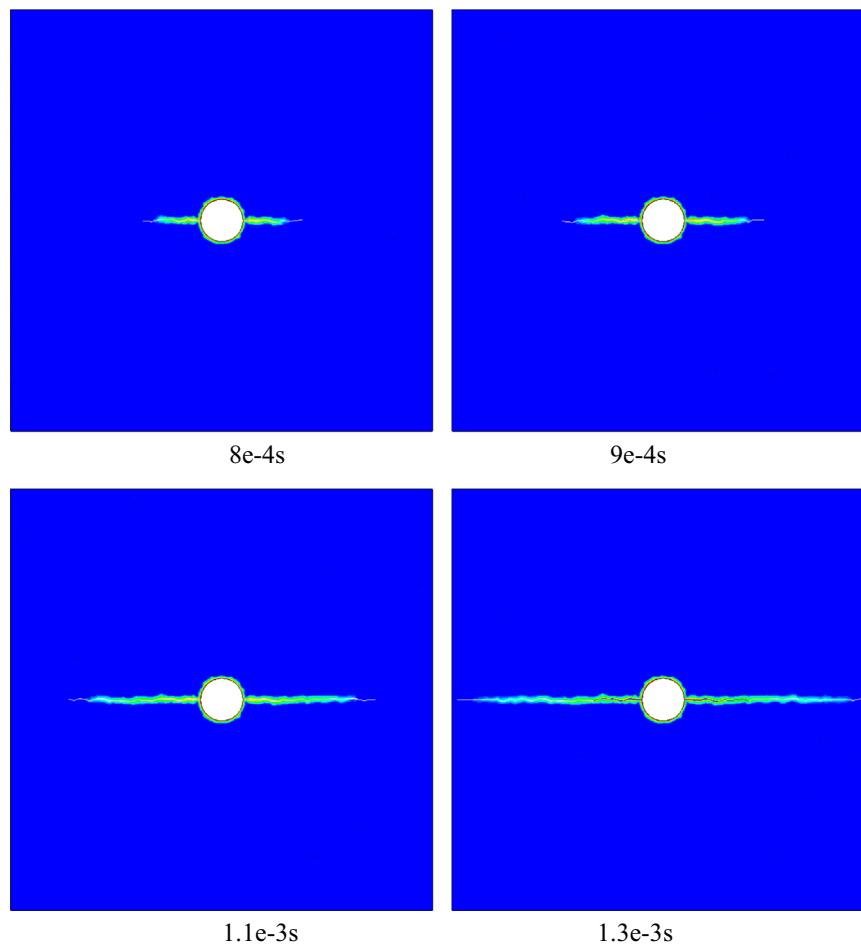


Fig. 12. (continued)

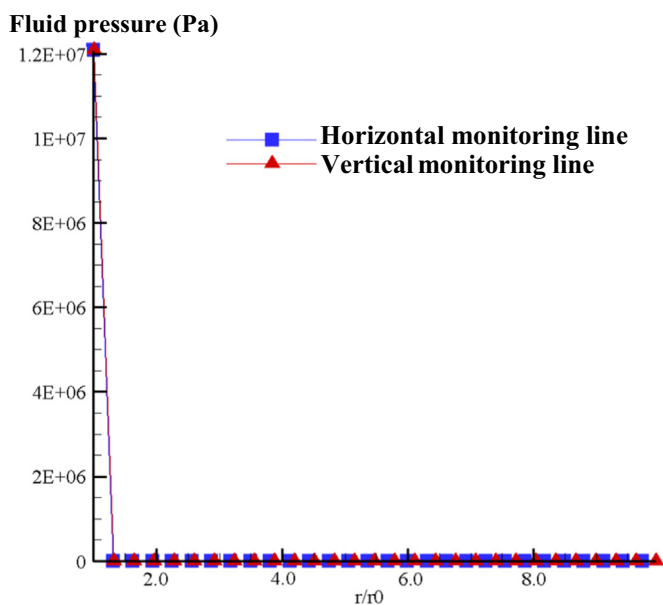


Fig. 13. Fluid pressure distribution before crack generation: high viscosity fluid used.

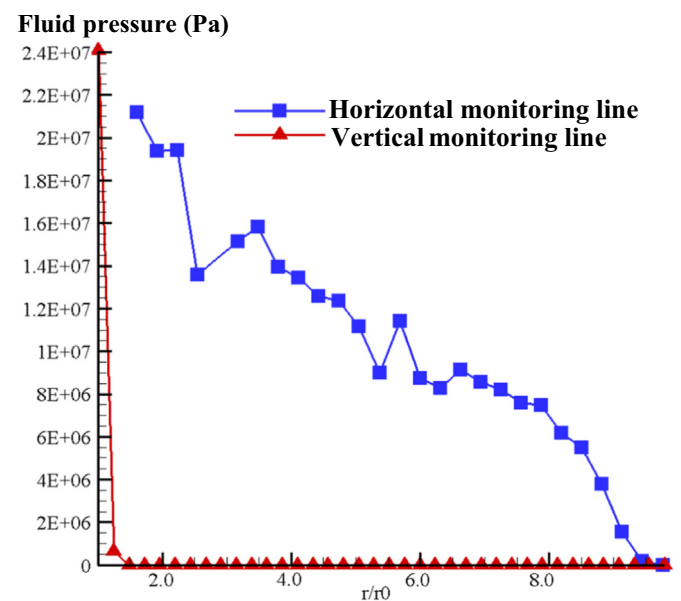


Fig. 14. Fluid pressure distribution at time step 130,000: high viscosity fluid used.

linearly. Because the fluid viscosity is large, its permeation into rock is almost impossible, and the accumulated amount of fluid within the borehole increases linearly. Despite the fact that after 4.5e-4s, the sample has fractured, the fluid pressure within the

borehole does not decrease, but continues to increase, even after the fracture has extended a certain length, as shown at 9e-4s in Fig. 12. Because a great deal of fluid is stored in the borehole rather than inflows to the fracture, until 1.4e-3s. By then, the fluid pressure is slightly lower and plateauing, as shown in Fig. 15(a),

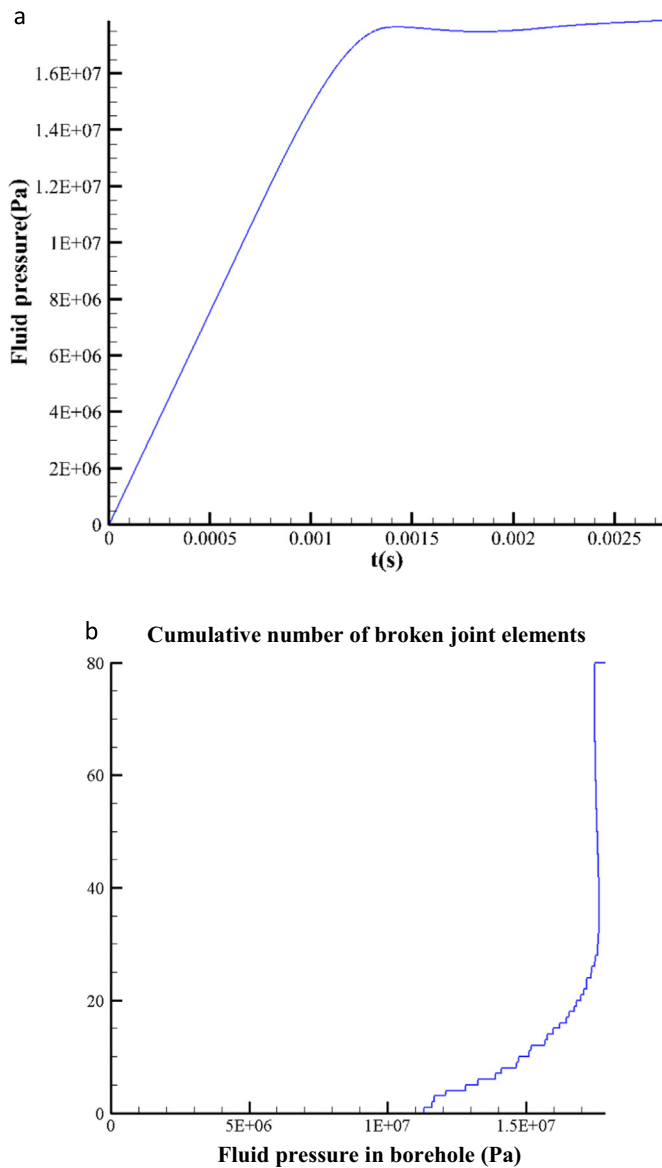


Fig. 15. (a) Fluid pressure - time step curve: high viscosity fluid injection; and (b) cumulative number of broken joint elements-fluid pressure curve: high viscosity fluid injection.

Table 2
Initiation and breakdown pressures.

	Initiation pressure (MPa)	Breakdown pressure (MPa)	Theoretical value of initiation pressure (MPa)
Low viscosity fluid (0.1 mPa)	8.1	8.4	11.0
High viscosity fluid (100 mPa)	11.5	17.5	11.0

and the breakdown pressure, 17.5 MPa, is reached. The fluid pressure in the borehole remains stable – very different from the low viscosity injection, where the fluid pressure drops significantly after passing the critical point.

In addition, according to Fig. 15(b), when the fluid pressure within the borehole is 11.5 MPa (but the breakdown pressure is 17.5 MPa, larger than the initiation pressure), the rock starts to crack. The pressure of 11.5 MPa is close to the theoretical value of the initiation pressure

$$P_b = 3\sigma_h - \sigma_H + f_t = 11 \text{ MPa.} \quad (22)$$

The result is different from a low viscosity injection. The reason for the calculated initiation pressure to be close to the theoretical initiation pressure is because before cracking the highly viscous fluid is almost unlikely to infiltrate into the rock; therefore, the fluid pressure only acts on the borehole surface – the same as the load condition from which the theoretical formula is derived. This conclusion is consistent with the experimental data^{59,60} and verifies the correctness of FDEM-flow.

In summary, as shown in Table 2, when the low viscosity fluid is used, the initiation pressure is lower than the value from the theoretical formula, and the breakdown pressure is close to the initiation pressure. When high viscosity fluid is injected, the initiation pressure is close to the value from the theoretical formula, and the breakdown pressure is much larger than the initiation pressure. Meanwhile, when a high viscosity fluid is injected, the initiation pressure and breakdown pressure are larger than both the initiation pressures and breakdown pressure seen with low viscosity fluid injection.

9. Conclusions

From the preceding analysis, the FDEM-flow model is able to consider porous media flow in simulating hydraulic fracturing, and the following conclusions are drawn.

When a low viscosity fluid is used to inject, before crack initiation, the zone of fluid pressure distribution expands with time. The permeability coefficient is large in the direction of the major principal stress; however, before crack initiation, the effect of *in situ* stress on the permeability coefficient is small. After initiation, the permeability along the direction of crack propagation is greater than that normal to it. The *in situ* stress exerts a significant influence on the permeability.

When a high viscosity fluid is used, before crack initiation, the directional nature of permeability is not obvious under the effects of *in situ* stress, and that the effect of *in situ* stress on the permeability is limited. After crack initiation, the permeability along the direction of crack propagation is much larger than that normal to it. At this time, the *in situ* stress exerts a significant influence on the permeability coefficient.

When a low viscosity fluid is used, firstly, the fluid pressure within the borehole increases non-linearly, then it drops significantly. When a high viscosity fluid is used, firstly, the fluid pressure within the borehole increases linearly, then remains unchanged.

When a low viscosity fluid is injected, the fluid seeps into the cracks immediately upon crack initiation. Under the effects of pore pressure, the subsequent propagation of the crack is relatively easy. When a high viscosity fluid is injected, The fluid infiltrates into the extended fractures slowly. The subsequent crack propagation is relatively difficult.

When a low viscosity fluid injection is applied, the initiation pressure is lower than the value from the theoretical formula, and the breakdown pressure is close to the initiation pressure. When a high viscosity fluid injection is applied, the initiation pressure is relatively close to the value from the theoretical formula, and the breakdown pressure is much larger than initiation pressure. Meanwhile, the initiation pressure and breakdown pressure are larger than those from the low viscosity fluid injection.

The cracks extend along the direction of the major principal stress, suggesting that the effects of *in situ* stress are properly considered.

The initiation pressure predicted by the theoretical formula is correct only when the permeability of the rock is low, or the

viscosity of fluid is large. Otherwise, the formula is not appropriate. In general, the initiation pressure assessed by the theoretical formula is larger than the actual initiation pressure.

In conclusion, with FDEM-flow, we can reproduce the stress and fluid pressure distributions, and crack propagation paths. We will consider the extension of this model to three-dimensional problems in future.

Acknowledgements

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